SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 18-Apr-2008 **Revision Date** 09-Feb-2024 **Revision Number** 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Isobutyric anhydride**
Cat No. :	**L13240**
Synonyms	2-Methylpropionic anhydride
CAS No	97-72-3
EC No	202-603-6
Molecular Formula	C8 H14 O3
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

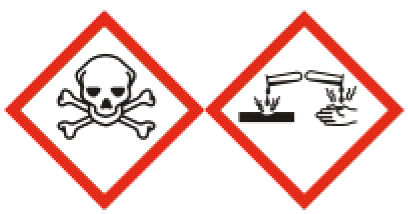
Acute dermal toxicity	Category 3 (H311)
Acute Inhalation Toxicity - Vapors	Category 3 (H331)
Skin Corrosion/Irritation	Category 1 B (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word **Danger**

Hazard Statements

H314 - Causes severe skin burns and eye damage
H311 + H331 - Toxic in contact with skin or if inhaled
Combustible liquid

Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

2.3. Other hazards

Decomposes in contact with water
Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates
This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Isobutyric anhydride	97-72-3	EEC No. 202-603-6	>95	Skin Corr. 1B (H314) Eye Dam. 1 (H318) Acute Tox. 3 (H331) Acute Tox. 3 (H311)
Propanoic acid	79-09-4	EEC No. 201-176-3	<=1	Flam. Liq. 3 (H226) Met. Corr. 1 (H290) Skin Corr. 1B (H314) Eye Corr. 1 (H318) STOT SE 3 (H335)
Acetic acid	64-19-7	200-580-7	<=0.5	Flam. Liq. 3 (H226) Skin Corr. 1A (H314) Eye Dam. 1 (H318)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Propanoic acid	Eye Irrit. 2 (H319) :: 10%<=C<25% Skin Corr. 1B (H314) :: C>=25% Skin Irrit. 2 (H315) :: 10%<=C<25% STOT SE 3 (H335) :: C>=10%	-	-
Acetic acid	Skin Corr. 1A (H314) :: C>=90% Skin Corr. 1B (H314) :: 25%<=C<90% Eye Irrit. 2 (H319) :: 10%<=C<25% Skin Irrit. 2 (H315) :: 10%<=C<25%	-	-

REACH registration number	-	
Components	**Reach Registration Number**	
Isobutyric anhydride	01-2119902390-51	

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation Remove to fresh air. If breathing is difficult, give oxygen. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes burns by all exposure routes. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Carbon dioxide (CO_2). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons
Water.

5.2. Special hazards arising from the substance or mixture

Combustible material. Flammable. Vapors may form explosive mixtures with air. Containers may explode when heated.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

6.2. Environmental precautions

See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not let this chemical enter the environment. Remove all sources of ignition.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep away from open flames, hot surfaces and sources of ignition. Avoid contact with skin, eyes or clothing. Avoid breathing dust/fume/gas/mist/vapors/spray. Avoid ingestion and inhalation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) — Storage Class/LGK 6.1C

Switzerland - Storage of hazardous substances — Storage class - SC 6.1
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Propanoic acid	TWA: 10 ppm (8h) TWA: 31 mg/m³ (8h) STEL: 20 ppm (15min) STEL: 62 mg/m³ (15min)	STEL: 15 ppm 15 min STEL: 46 mg/m³ 15 min TWA: 10 ppm 8 hr TWA: 31 mg/m³ 8 hr	TWA / VME: 10 ppm (8 heures). indicative limit TWA / VME: 31 mg/m³ (8 heures). indicative limit STEL / VLCT: 20 ppm. indicative limit STEL / VLCT: 62 mg/m³. indicative limit	TWA: 10 ppm 8 uren TWA: 31 mg/m³ 8 uren STEL: 20 ppm 15 minuten STEL: 62 mg/m³ 15 minuten	STEL / VLA-EC: 20 ppm (15 minutos). STEL / VLA-EC: 62 mg/m³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 31 mg/m³ (8 horas)
Acetic acid	TWA: 25 mg/m³ (8h) TWA: 10 ppm (8h) STEL: 50 mg/m³ (15min) STEL: 20 ppm (15min)	STEL: 37 mg/m³ STEL: 15 ppm TWA: 10 ppm TWA: 25 mg/m³	TWA / VME: 10 ppm (8 heures). TWA / VME: 25 mg/m³ (8 heures). STEL / VLCT: 20 ppm. indicative limit STEL / VLCT: 50 mg/m³. indicative limit	TWA: 10 ppm 8 uren TWA: 25 mg/m³ 8 uren STEL: 15 ppm 15 minuten STEL: 38 mg/m³ 15 minuten	STEL / VLA-EC: 20 ppm (15 minutos). STEL / VLA-EC: 50 mg/m³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 25 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Propanoic acid	TWA: 10 ppm 8 ore. Time Weighted Average TWA: 31 mg/m³ 8 ore. Time Weighted Average STEL: 20 ppm 15 minuti. Short-term STEL: 62 mg/m³ 15 minuti. Short-term	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 31 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK TWA: 31 mg/m³ (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 62 mg/m³	STEL: 20 ppm 15 minutos STEL: 62 mg/m³ 15 minutos TWA: 10 ppm 8 horas TWA: 31 mg/m³ 8 horas	STEL: 62 mg/m³ 15 minuten TWA: 31 mg/m³ 8 uren	TWA: 10 ppm 8 tunteina TWA: 31 mg/m³ 8 tunteina STEL: 20 ppm 15 minuutteina STEL: 61 mg/m³ 15 minuutteina
Acetic acid	TWA: 25 ppm 8 ore. Time Weighted Average TWA: 10 mg/m³ 8 ore. Time Weighted Average STEL: 50 mg/m³ 15 minuti. Short-term STEL: 20 ppm 15 minuti. Short-term	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 25 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK TWA: 25 mg/m³ (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 50 mg/m³	STEL: 20 ppm 15 minutos STEL: 50 mg/m³ 15 minutos TWA: 10 ppm 8 horas TWA: 25 mg/m³ 8 horas	MAC-TGG 25 mg/m³	TWA: 5 ppm 8 tunteina TWA: 13 mg/m³ 8 tunteina STEL: 10 ppm 15 minuutteina STEL: 25 mg/m³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Propanoic acid	MAK-KZGW: 20 ppm 15 Minuten MAK-KZGW: 62 mg/m³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 31 mg/m³ 8 Stunden	TWA: 10 ppm 8 timer TWA: 31 mg/m³ 8 timer STEL: 62 mg/m³ 15 minutter STEL: 20 ppm 15 minutter	STEL: 20 ppm 15 Minuten STEL: 60 mg/m³ 15 Minuten TWA: 10 ppm 8 Stunden TWA: 30 mg/m³ 8 Stunden	STEL: 45 mg/m³ 15 minutach TWA: 30 mg/m³ 8 godzinach	TWA: 10 ppm 8 timer TWA: 30 mg/m³ 8 timer STEL: 20 ppm 15 minutter. value calculated STEL: 45 mg/m³ 15 minutter. value calculated
Acetic acid	MAK-KZGW: 20 ppm 15 Minuten MAK-KZGW: 50 mg/m³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 25 mg/m³ 8 Stunden	TWA: 10 ppm 8 timer TWA: 25 mg/m³ 8 timer STEL: 50 mg/m³ 15 minutter STEL: 20 ppm 15 minutter	STEL: 20 ppm 15 Minuten STEL: 50 mg/m³ 15 Minuten TWA: 10 ppm 8 Stunden TWA: 25 mg/m³ 8 Stunden	STEL: 50 mg/m³ 15 minutach TWA: 25 mg/m³ 8 godzinach	TWA: 10 ppm 8 timer TWA: 25 mg/m³ 8 timer STEL: 20 ppm 15 minutter. value from the regulation STEL: 50 mg/m³ 15 minutter. value from the regulation

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Propanoic acid	TWA: 10 ppm TWA: 31.0 mg/m³ STEL : 20 ppm STEL : 62.0 mg/m³	TWA-GVI: 10 ppm 8 satima. TWA-GVI: 31 mg/m³ 8 satima. STEL-KGVI: 20 ppm 15 minutama. STEL-KGVI: 62 mg/m³ 15 minutama.	TWA: 10 ppm 8 hr. TWA: 31 mg/m³ 8 hr. STEL: 20 ppm 15 min STEL: 62 mg/m³ 15 min	STEL: 20 ppm STEL: 62 mg/m³ TWA: 10 ppm TWA: 31 mg/m³	TWA: 30 mg/m³ 8 hodinách. Ceiling: 60 mg/m³
Acetic acid	TWA: 25 mg/m³ TWA: 10 ppm STEL : 50 mg/m³ STEL : 20 ppm	TWA-GVI: 10 ppm 8 satima. TWA-GVI: 25 mg/m³ 8 satima. STEL-KGVI: 20 ppm 15 minutama. STEL-KGVI: 50 mg/m³ 15 minutama.	TWA: 20 ppm 8 hr. TWA: 50 mg/m³ 8 hr. STEL: 20 ppm 15 min STEL: 50 mg/m³ 15 min	STEL: 50 mg/m³ STEL: 20 ppm TWA: 10 ppm TWA: 25 mg/m³	TWA: 25 mg/m³ 8 hodinách. Ceiling: 50 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Propanoic acid	TWA: 10 ppm 8 tundides. TWA: 30 mg/m³ 8 tundides. STEL: 20 ppm 15 minutites.	TWA: 10 ppm 8 hr TWA: 31 mg/m³ 8 hr STEL: 20 ppm 15 min STEL: 62 mg/m³ 15 min	STEL: 20 ppm STEL: 60 mg/m³ TWA: 10 ppm TWA: 30 mg/m³	STEL: 62 mg/m³ 15 percekben. CK TWA: 31 mg/m³ 8 órában. AK	STEL: 20 ppm STEL: 62 mg/m³ TWA: 10 ppm 8 klukkustundum. TWA: 31 mg/m³ 8 klukkustundum.

	STEL: 62 mg/m^3 15 minutites.				
Acetic acid	TWA: 10 ppm 8 tundides. TWA: 25 mg/m^3 8 tundides. STEL: 10 ppm 15 minutites. STEL: 25 mg/m^3 15 minutites.	TWA: 25 mg/m^3 8 hr TWA: 10 ppm 8 hr STEL: 50 mg/m^3 15 min STEL: 20 ppm 15 min	STEL: 15 ppm STEL: 37 mg/m^3 TWA: 10 ppm TWA: 25 mg/m^3	STEL: 50 mg/m^3 15 percekben. CK TWA: 25 mg/m^3 8 órában. AK	STEL: 20 ppm STEL: 50 mg/m^3 TWA: 10 ppm 8 klukkustundum. TWA: 25 mg/m^3 8 klukkustundum.

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Propanoic acid	STEL: 20 ppm STEL: 62 mg/m^3 TWA: 10 ppm TWA: 31 mg/m^3	TWA: 10 ppm IPRD TWA: 31 mg/m^3 IPRD STEL: 20 ppm STEL: 62 mg/m^3	TWA: 10 ppm 8 Stunden TWA: 31 mg/m^3 8 Stunden STEL: 20 ppm 15 Minuten STEL: 62 mg/m^3 15 Minuten	TWA: 10 ppm TWA: 31 mg/m^3 STEL: 20 ppm 15 minuti STEL: 62 mg/m^3 15 minuti	TWA: 10 ppm 8 ore TWA: 31 mg/m^3 8 ore STEL: 20 ppm 15 minute STEL: 62 mg/m^3 15 minute
Acetic acid	STEL: 50 mg/m^3 STEL: 20 ppm TWA: 10 ppm TWA: 25 mg/m^3	TWA: 10 ppm IPRD TWA: 25 mg/m^3 IPRD STEL: 50 mg/m^3 STEL: 20 ppm	TWA: 10 ppm 8 Stunden TWA: 25 mg/m^3 8 Stunden STEL: 50 mg/m^3 15 Minuten STEL: 20 ppm 15 Minuten	TWA: 10 ppm TWA: 25 mg/m^3 STEL: 20 ppm 15 minuti STEL: 50 mg/m^3 15 minuti	TWA: 10 ppm 8 ore TWA: 25 mg/m^3 8 ore STEL: 20 ppm 15 minute STEL: 50 mg/m^3 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Propanoic acid	MAC: 20 mg/m^3	Ceiling: 62 mg/m^3 TWA: 10 ppm TWA: 31 mg/m^3	TWA: 10 ppm 8 urah TWA: 31 mg/m^3 8 urah STEL: 20 ppm 15 minutah STEL: 62 mg/m^3 15 minutah	Binding STEL: 20 ppm 15 minuter Binding STEL: 62 mg/m^3 15 minuter TLV: 10 ppm 8 timmar. NGV TLV: 30 mg/m^3 8 timmar. NGV	TWA: 10 ppm 8 saat TWA: 31 mg/m^3 8 saat STEL: 20 ppm 15 dakika STEL: 62 mg/m^3 15 dakika
Acetic acid	Skin notation MAC: 5 mg/m^3	Ceiling: 50 mg/m^3 TWA: 10 ppm TWA: 25 mg/m^3	TWA: 10 ppm 8 urah TWA: 25 mg/m^3 8 urah STEL: 50 mg/m^3 15 minutah STEL: 20 ppm 15 minutah	Binding STEL: 10 ppm 15 minuter Binding STEL: 25 mg/m^3 15 minuter TLV: 5 ppm 8 timmar. NGV TLV: 13 mg/m^3 8 timmar. NGV	TWA: 10 ppm 8 saat TWA: 25 mg/m^3 8 saat

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

Component	Acute effects local	Acute effects	Chronic effects local	Chronic effects

	(Dermal)	systemic (Dermal)	(Dermal)	systemic (Dermal)
Isobutyric anhydride 97-72-3 (>95)				DNEL = 3.75mg/kg bw/day
Propanoic acid 79-09-4 (<=1)				DNEL = 20.9mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Isobutyric anhydride 97-72-3 (>95)				DNEL = 184mg/m³
Propanoic acid 79-09-4 (<=1)	DNEL = 62mg/m³		DNEL = 31mg/m³	DNEL = 73mg/m³
Acetic acid 64-19-7 (<=0.5)	DNEL = 25mg/m³		DNEL = 25mg/m³	

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Isobutyric anhydride 97-72-3 (>95)	PNEC = 0.026mg/L	PNEC = 0.364mg/kg sediment dw	PNEC = 0.26mg/L	PNEC = 19mg/L	PNEC = 0.0462mg/kg soil dw
Propanoic acid 79-09-4 (<=1)	PNEC = 0.5mg/L	PNEC = 1.86mg/kg sediment dw	PNEC = 5mg/L	PNEC = 5mg/L	PNEC = 0.1258mg/kg soil dw
Acetic acid 64-19-7 (<=0.5)	PNEC = 3.058mg/L	PNEC = 11.36mg/kg sediment dw	PNEC = 30.58mg/L	PNEC = 85mg/L	PNEC = 0.47mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Isobutyric anhydride 97-72-3 (>95)	PNEC = 0.0026mg/L	PNEC = 0.0363mg/kg sediment dw			
Propanoic acid 79-09-4 (<=1)	PNEC = 0.05mg/L	PNEC = 0.186mg/kg sediment dw			
Acetic acid 64-19-7 (<=0.5)	PNEC = 0.3058mg/L	PNEC = 1.136mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures
Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers	-		(minimum requirement)

Neoprene Natural rubber PVC	recommendations	EN 374

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	pungent	
Odor Threshold	No data available	
Melting Point/Range	-53 °C / -63.4 °F	
Softening Point	No data available	
Boiling Point/Range	182 °C / 359.6 °F	@ 760 mmHg
Flammability (liquid)	Combustible liquid	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	**Lower** 1.09 Vol% **Upper** 7.7 Vol%	
Flash Point	66.4 °C / 151.5 °F	**Method -** No information available
Autoignition Temperature	329 °C / 624.2 °F	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	1.18 cP (25°C)	
Water Solubility	Decomposes	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Isobutyric anhydride	1.1	
Propanoic acid	0.33	
Acetic acid	-0.2	
Vapor Pressure	0.7 mbar @ 20 °C	
Density / Specific Gravity	0.954	
Bulk Density	Not applicable	Liquid

Vapor Density	No information available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula C8 H14 O3
Molecular Weight 158.2
Explosive Properties explosive air/vapour mixtures possible

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Moisture sensitive.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions No information available.

10.4. Conditions to avoid

Heat, flames and sparks. Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Strong oxidizing agents. Strong bases. Alcohols.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information No acute toxicity information is available for this product

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Category 3
Inhalation Category 3

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Isobutyric anhydride	2230 mg/kg (Rat)	474 mg/kg (Rabbit)	4.21 mg/L/7h (Rat)
Propanoic acid	LD50 = 3455 mg/kg (Rat)	LD50 = 3235 mg/kg (Rabbit)	LC50 = > 19.7 mg/l (Rat) 1 h
Acetic acid	3310 mg/kg (Rat)	-	> 40 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory	Based on available data, the classification criteria are not met
Skin	Based on available data, the classification criteria are not met
(e) germ cell mutagenicity;	Based on available data, the classification criteria are not met
(f) carcinogenicity;	Based on available data, the classification criteria are not met There are no known carcinogenic chemicals in this product
(g) reproductive toxicity;	Based on available data, the classification criteria are not met
(h) STOT-single exposure;	Based on available data, the classification criteria are not met
(i) STOT-repeated exposure;	Based on available data, the classification criteria are not met
Target Organs	None known.
(j) aspiration hazard;	Based on available data, the classification criteria are not met
Other Adverse Effects	The toxicological properties have not been fully investigated.
Symptoms / effects,both acute and delayed	Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Do not empty into drains. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Isobutyric anhydride	Leuciscus idus: LC50= 146 mg/L/96h	EC50= 51.25 mg/L/48h	
Propanoic acid	LC50: = 51 mg/L, 96h static (Oncorhynchus mykiss) LC50: 73 - 99.7 mg/L, 96h static (Lepomis macrochirus) LC50: > 1 mg/L, 96h static (Pimephales promelas)		EC50: = 45.8 mg/L, 72h (Desmodesmus subspicatus) EC50: = 43 mg/L, 96h (Desmodesmus subspicatus)
Acetic acid	Pimephales promelas: LC50 = 88 mg/L/96h Lepomis macrochirus: LC50 = 75 mg/L/96h	EC50 = 95 mg/L/24h	-

Component	Microtox	M-Factor

Propanoic acid	EC50 = 59.6 mg/L 17 h	
Acetic acid	Photobacterium phosphoreum: EC50 = 8.8 mg/L/15 min Photobacterium phosphoreum: EC50 = 8.8 mg/L/25 min Photobacterium phosphoreum: EC50 = 8.8 mg/L/5 min	

12.2. Persistence and degradability

Persistence Persistence is unlikely, based on information available.
Degradability Decomposes in contact with water.
Degradation in sewage treatment plant No information available. Decomposes in contact with water.

12.3. Bioaccumulative potential Product does not bioaccumulate due to reaction with water

Component	log Pow	Bioconcentration factor (BCF)
Isobutyric anhydride	1.1	No data available
Propanoic acid	0.33	No data available
Acetic acid	-0.2	No data available

12.4. Mobility in soil Decomposes in contact with water . Is not likely mobile in the environment.

12.5. Results of PBT and vPvB assessment Decomposes in contact with water. Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties
Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects
Persistent Organic Pollutant This product does not contain any known or suspected substance
Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	(ISOBUTYRIC ANHYDRIDE)
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

ADR

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	(ISOBUTYRIC ANHYDRIDE)
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

IATA

14.1. UN number	UN2922
14.2. UN proper shipping name	Corrosive liquid, toxic, n.o.s.
Technical Shipping Name	(ISOBUTYRIC ANHYDRIDE)
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	6.1
14.4. Packing group	II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Isobutyric anhydride	97-72-3	202-603-6	-	-	X	X	KE-24876	X	X
Propanoic acid	79-09-4	201-176-3	-	-	X	X	KE-29352	X	X
Acetic acid	64-19-7	200-580-7	-	-	X	X	X	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Isobutyric anhydride	97-72-3	X	ACTIVE	X	-	X	X	X
Propanoic acid	79-09-4	X	ACTIVE	X	-	X	X	X
Acetic acid	64-19-7	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Isobutyric anhydride	97-72-3	-	-	-
Propanoic acid	79-09-4	-	Use restricted. See item 75. (see link for restriction details)	-
Acetic acid	64-19-7	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Isobutyric anhydride	97-72-3	Not applicable	Not applicable
Propanoic acid	79-09-4	Not applicable	Not applicable
Acetic acid	64-19-7	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Isobutyric anhydride	WGK1	
Propanoic acid	WGK1	
Acetic acid	WGK1	Class II : 0.10 g/m³ (Massenkonzentration)

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent

	substances preparation (SR 814.81)		**Procedure**
Propanoic acid 79-09-4 (<=1)	Prohibited and Restricted Substances		
Acetic acid 64-19-7 (<=0.5)	Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H226 - Flammable liquid and vapor
H311 - Toxic in contact with skin
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H331 - Toxic if inhaled

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

Prepared By	Health, Safety and Environmental Department
Creation Date	18-Apr-2008
Revision Date	09-Feb-2024
Revision Summary	New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet